MANIPULATED IMAGE DETECTION AND SEGMENTATION

Deep Learning Semester Project Report

BSCS VII

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Abstract
Digital image manipulation has become a growing concern in the modern era of social media, journalism, and legal evidence handling. With the advancement of generative AI and powerful editing tools, forged images can be created and shared effortlessly, raising critical questions about visual authenticity and digital trust. This project presents a deep learning–based approach for detecting and localizing manipulated regions in digital images using the diverse DEFACTO Image Forgery Dataset, covering Splicing, Inpainting, Copy-Move, and Face manipulations.

The developed system implements a dual-stage framework: a ResNet-50 Convolutional Neural Network (CNN) for multi-class classification to identify the type of forgery, and a DeepLabV3+ model with a ResNet-50 backbone for precise segmentation of manipulated regions. The dataset underwent rigorous preprocessing, including resizing, normalization, and augmentation, to ensure robust model generalization.

Experimental results demonstrate the system's effectiveness, with the classification model achieving an Accuracy of 90.2% and an F1-Score of 90.5%. The segmentation model demonstrated capability in localizing tampered areas, achieving an Intersection over Union (IoU) of 0.622 and a Dice Coefficient of 0.669. By combining high-accuracy classification with pixel-level localization, this project contributes a viable solution to digital forensics, offering an explainable method for verifying image integrity.

1.0 Introduction

In today's digital era, image manipulation has become increasingly sophisticated due to the rapid advancement of image editing tools and generative AI technologies [1][2]. While such tools enable creativity and artistic expression, they also pose serious threats to digital integrity, online authenticity, and media credibility. Images altered to mislead or deceive are frequently used in misinformation campaigns, legal disputes, and digital forensics, making forgery detection a crucial research area in computer vision [3][4].

Traditional image forgery detection methods, based on handcrafted features such as noise inconsistencies or JPEG artifacts, have shown limited success in identifying complex manipulations [5]. With the emergence of Deep Learning, Convolutional Neural Networks (CNNs) have demonstrated remarkable capability in automatically learning discriminative spatial features from large-scale image data, enabling higher accuracy in classifying manipulated images [6][7]. However, while most existing methods can detect whether an image has been tampered with, localizing the exact manipulated regions remains a major challenge [8].

To address this limitation, this project proposes a deep learning–based system that can both detect and localize image forgeries using the DEFACTO Dataset, a comprehensive benchmark for image manipulation detection [9]. The proposed approach integrates a CNN-based classification model to determine whether an image is authentic or forged [6], and a U-Net (or DeepLabV3+) based segmentation model to generate precise binary masks highlighting tampered regions [10][11]. By combining detection and localization in a unified workflow, this study aims to contribute to the field of digital forensics and image authenticity verification with improved accuracy and interpretability [12][13].

2.0 Problem Identification

2.1 Background

With the exponential growth of digital content, image manipulation has emerged as a serious concern across social media, journalism, law enforcement, and digital forensics [1] [3]. Modern editing software such as Adobe Photoshop, GIMP, and AI-based generative models (e.g., GANs, diffusion models) can produce highly realistic forgeries that are nearly indistinguishable from authentic images to the human eye [2][4].

Manipulated images are widely used to spread misinformation, fabricate evidence, or alter public perception, posing threats to both individual reputation and societal trust in visual media [3][4].

Traditional forgery detection methods rely on handcrafted features such as color inconsistencies, noise patterns, or JPEG compression artifacts [5]. However, these methods often fail to detect complex and high-quality manipulations, particularly when forgeries involve semantic-level editing such as object removal, content swapping, or AI-generated enhancements [4].

With the advent of Deep Learning, particularly Convolutional Neural Networks (CNNs), automatic feature learning has enabled better performance in distinguishing between real and manipulated images [6][7]. Yet, most existing models still operate as binary classifiers - they can identify whether an image is fake but cannot localize or visualize the tampered areas [8].

This lack of spatial interpretability limits their practical utility in forensic and investigative applications where pinpointing the manipulated region is essential [12][13].

2.2 Research Questions

The core questions guiding this project are:

1. Detection – How can a deep learning model be designed to accurately distinguish between authentic and manipulated images using the DEFACTO dataset?
2. Localization – Can a segmentation network (e.g., U-Net or DeepLabV3+) effectively identify and highlight the precise regions that have been manipulated?
3. Performance & Generalization – How well does the proposed dual-model approach perform across various types of manipulations (e.g., copy-move, splicing, object removal, morphing), and can it generalize to unseen forgery types?
4. Interpretability – Can the integration of classification and segmentation improve explainability in image forgery detection systems?

2.3 Hypothesis

It is hypothesized that:

1. Finetuning a pretrained classification model can learn discriminative features that differentiate authentic and manipulated images with high accuracy.
2. A segmentation model DeepLabV3+, trained with Binary Cross Entropy (BCE) and Dice Loss, can accurately localize manipulated regions by generating precise binary masks.
3. Combining both models within a unified detection–localization pipeline will enhance both accuracy and interpretability, outperforming standalone classification or segmentation approaches.

2.4 Contributions of the Project

This project aims to make the following contributions to the field of image forensics and deep learning–based forgery detection:

1. **Dual-Stage Detection Framework:**

Development of a two-stage deep learning pipeline that integrates image-level classification and pixel-level segmentation for comprehensive forgery analysis.

2. **Localization of Manipulated Regions:**

Implementation of a segmentation network capable of visualizing tampered regions, providing spatial interpretability that enhances forensic trust and usability.

3. Use of a Diverse Benchmark Dataset (DEFACTO):

Leveraging the DEFACTO dataset, which includes multiple manipulation types and groundtruth masks, to train and evaluate models with diverse and realistic examples.

4. Comprehensive Evaluation:

Detailed analysis using multiple metrics, Accuracy, Precision, Recall, F1-Score, IoU, Dice Coefficient - to ensure balanced and reliable assessment of both detection and localization performance.

5. Practical Application for Digital Forensics:

Providing a prototype that can serve as a foundational system for media verification tools, authenticity checks, and forensic analysis, contributing to digital trust and misinformation mitigation.

3.0 Objectives

The primary objectives of this project are as follows:

1. To design a classification model that can distinguish between authentic and forged images with high accuracy.
2. To implement a segmentation network (such as DeepLabV3+) to generate precise binary masks that highlight manipulated areas.
3. To evaluate and compare the model's performance using quantitative metrics such as Accuracy, Precision, Recall, F1-Score, IoU, and Dice Coefficient.
4. To analyze qualitative results through visual comparison of predicted masks and ground truth masks for different forgery types.

4.0 Dataset Discussion

The proposed system utilizes the DEFACTO (Deepfake and Image Forgery Detection) Dataset, which is a large-scale benchmark dataset specifically designed for training and evaluating deep learning models in image manipulation detection and localization tasks [16]. The DEFACTO dataset includes a diverse collection of images representing various types of manipulations, such as copy-move, splicing, object removal, and morphing, making it well-suited for developing robust and generalizable models [4][11][16].

Each manipulated image in the dataset is accompanied by a corresponding ground-truth mask that precisely highlights the tampered regions, providing essential pixel-level supervision for segmentation models like U-Net or DeepLabV3+ [6][8][10]. These masks improve the model's ability to learn localized manipulation patterns and enhance interpretability. The dataset also includes authentic (unaltered) images, ensuring balanced and unbiased training of the classification network [16].

There are four DEFACTO datasets being used defactosplicing, defactoinpainting, defactocopymove and defactoface respectively, each contributing to detecting a specific type of manipulation being performed in the image.

- **Defacto Splicing**

The type of manipulation where an object from another images is incorporated in an image is know as splicing manipulation. defacto-splicing dataset contains 106392 manipulated images sized 57 GB of data.

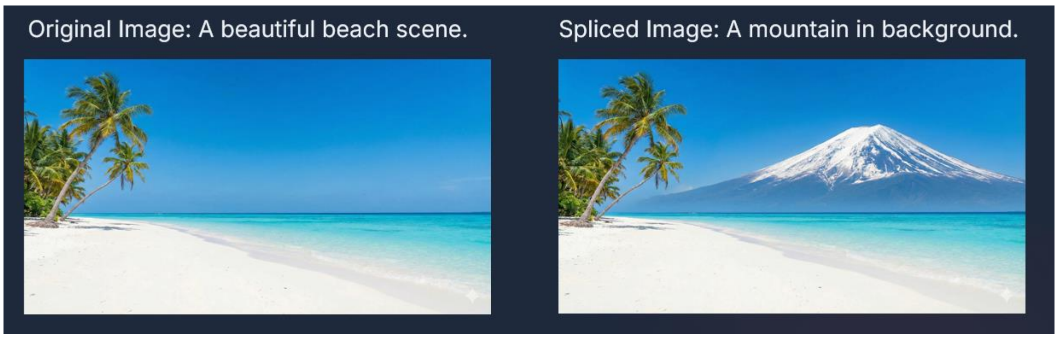

Figure 1: Splicing Manipulation

- **Defacto Inpainting**

The type of manipulation where an object is removed from the image is known as inpainting manipulation. defacto-inpainting contains 25001 samples of manipulated images, sized at 13 GB of data.

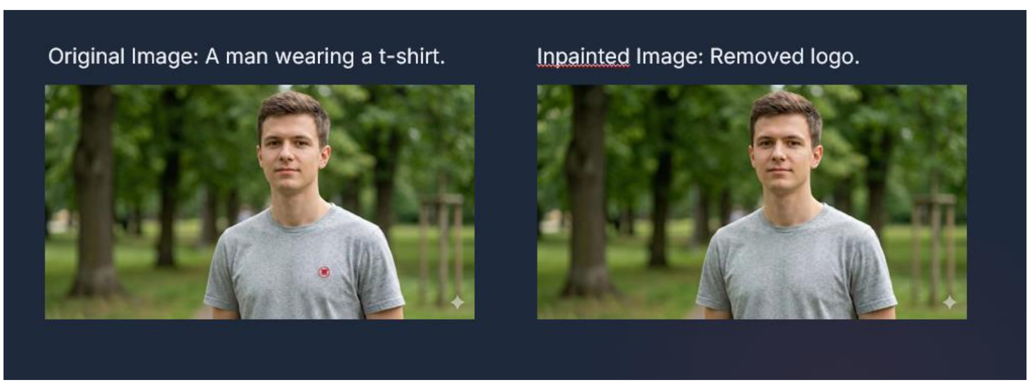

Figure 2: Inpainting Manipulation

- **Defacto Copymove**

When an object from the image is duplicated and placed in the same image is know as copymove manipulation. defacto-copymove contains 18194 manipulated images sized 10 GB of data.

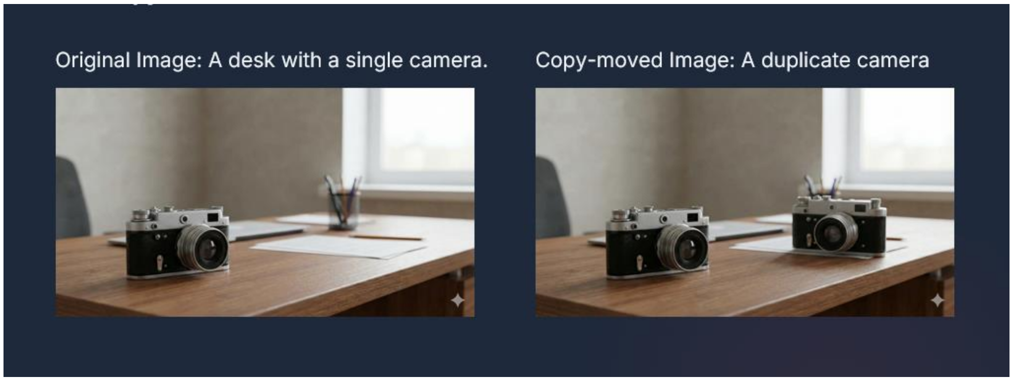

Figure 3: Copymove Manipulation

- **Defacto Face**

When manipulation is performed on face is known as face manipulation. It has two type, one is swapping where the face of one person is swapped with another and another type is morphing where the two faces are merged to create one face. Defacto-face contains 79600 manipulated (swapped + morphed combined) images sized 41 GB of data

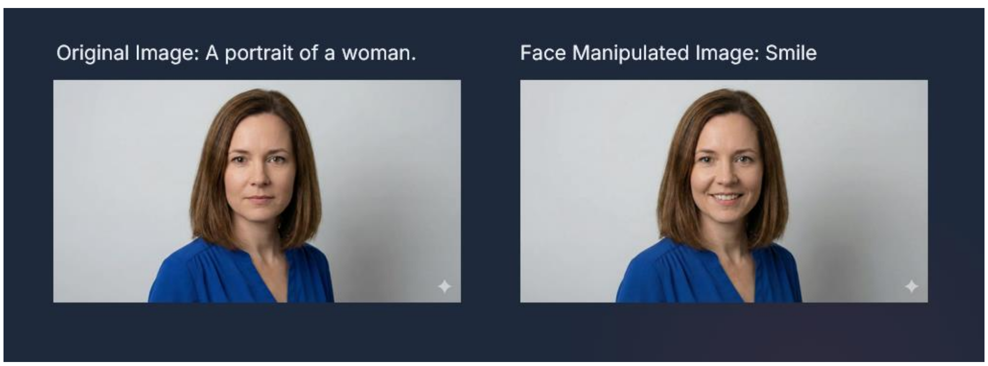

Figure 4: Face Manipulation

Dataset	Sample Size	Size	Description
DefactoSplicing	106392	57 GB	manipulation where an object from another images is incorporated in an image
DefactoInpainting	25001	13 GB	manipulation where an object is removed from the image
DefactoCopymove	18194	10 GB	Manipulation where an object is duplicated and placed in the same image
DefactoFace	79600	41 GB	manipulation performed on face

Table 1: Dataset Description

4.1 Dataset Preparation

Before training, the dataset undergoes the following preprocessing steps:

- **Resizing:** All images and masks are resized to a uniform dimension (256×256 pixels) to ensure compatibility with the network input layers.
- **Normalization:** Pixel values are scaled to the [0,1] range or standardized for stable model convergence.
- **Augmentation:** Techniques such as random rotation, flipping, and brightness variation are applied to improve model generalization and prevent overfitting.
- **Splitting:** The dataset is divided into 70% training, 15% validation, and 15% testing sets to ensure reliable evaluation and reproducibility.

4.2 Previous Benchmark Results

Several benchmark studies have evaluated the performance of deep learning models on the DEFACTO and related image manipulation datasets. These benchmarks provide a useful reference for setting performance expectations for this project.

- Bayar and Stamm (2016) [6] introduced one of the earliest CNN architectures specifically designed for detecting image manipulation. Their model achieved up to 97% classification accuracy on datasets containing splicing and copy-move forgeries. However, the method was limited to image-level classification and could not localize manipulated regions.
- Ronneberger et al. (2015) [7] proposed the U-Net architecture, which has since become a cornerstone for segmentation tasks. When adapted for image forgery localization, U-Net models typically achieve Intersection over Union (IoU) scores between 0.65–0.80 and Dice Coefficients of around 0.70–0.85, depending on dataset complexity and augmentation techniques.
- Chen et al. (2018) [8] improved upon U-Net with DeepLabV3+, introducing atrous convolution and an encoder-decoder structure that enhanced boundary precision. On manipulation datasets such as CASIA and DEFACTO, DeepLabV3+ has reported IoU values exceeding 0.80 and superior segmentation of fine-grained tampered regions compared to U-Net.
- Bappy et al. (2019) [13] proposed a hybrid LSTM and encoder-decoder approach, demonstrating F1-scores above 90% for binary classification and notable improvements in spatial localization. Their work highlights the advantage of combining spatial and sequential feature learning for manipulation detection.
- Korus and Huang (2017, 2019) [15][16], in developing the DEFACTO dataset, reported that baseline CNN models achieved classification accuracies of 92–95% and segmentation IoU scores around 0.70, establishing a strong benchmark for future models. However, they also noted that model generalization across manipulation types remains a key challenge, motivating the need for multiarchitecture and hybrid frameworks.

4.3 Rationale for Dataset Selection

The DEFACTO dataset was chosen for its variety of manipulation techniques, availability of ground-truth masks, and realistic image quality, which collectively make it a suitable choice for a deep learning model aimed at both forgery detection and localization. Its diversity across manipulation categories enables the proposed system to generalize effectively to unseen forgeries, thereby improving real-world applicability in digital forensics and image verification tasks.

5.0 Methodology

The project will follow a structured methodology with clearly defined steps:

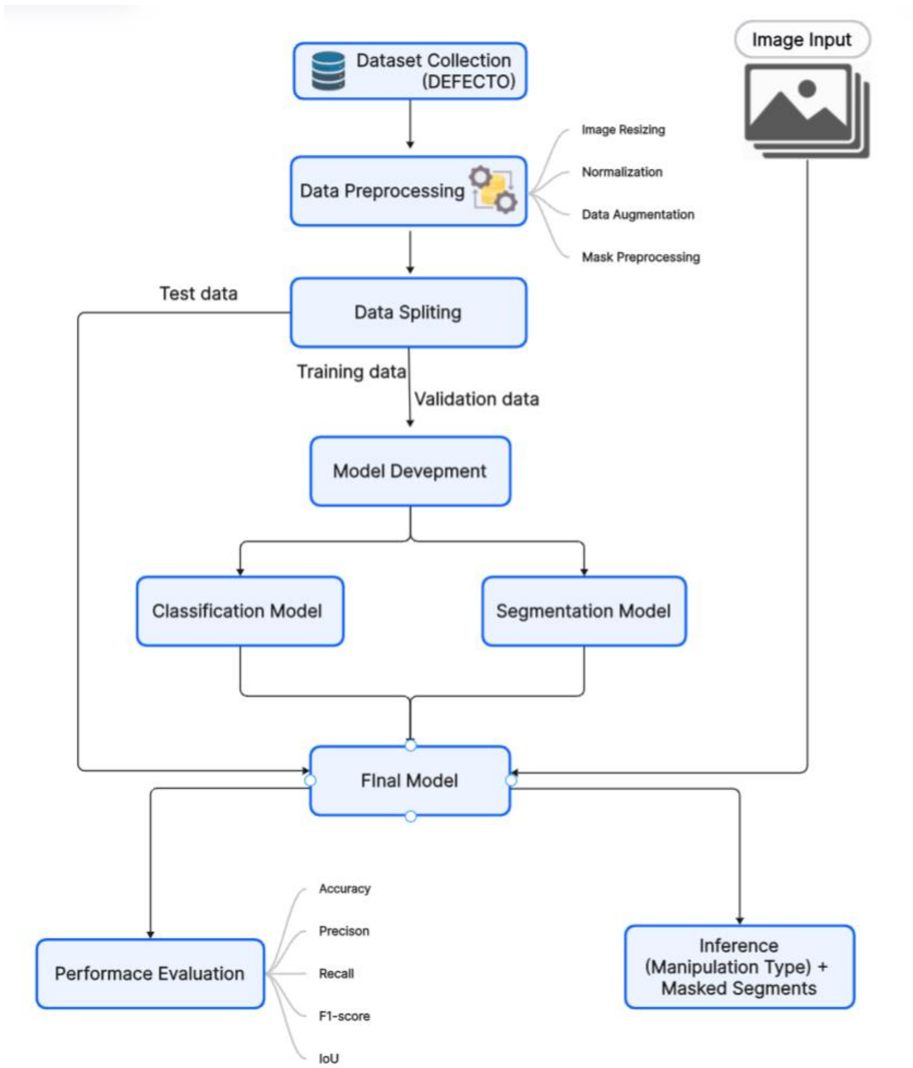

Figure 5: Proposed Methodology

This figure proposes a deep learning-based approach for detecting and localizing manipulated regions in images using the DEFACTO dataset. The overall workflow consists of dataset collection, preprocessing, data splitting, model development, and performance evaluation. The figure above shows the abstract model of the proposed methodology.

5.1 Model Overview

The main goal of this project is to build a system that can both detect whether an image is manipulated and localize the manipulated areas. The proposed approach includes two major models:

1. Classification Model – determines which type of manipulation is performed.
2. Segmentation Model – identifies and highlights the exact manipulated regions.

Both models are trained using the pre-processed data from the DEFACTO dataset. The complete process starts with dataset preparation and ends with performance evaluation using suitable metrics.

5.2 Architecture Description

The project uses two deep learning architectures for different tasks:

A. Classification Architecture

For the task of tampering-type identification, we utilized a ResNet-50 convolutional neural network, pre-trained on ImageNet, as the primary feature extractor.

- Input Layer: Accepts RGB images resized to 256 × 256 × 3.
- Residual Convolutional Blocks (ResNet-50 Backbone):
  - 50-layer deep residual architecture enabling high-level feature learning.
  - Skip connections preserve gradient flow and prevent gradient problems vanishing.
  - Extracts hierarchical visual representations ranging from low-level edges to complex semantic textures.
- Global Average Pooling: Reduces spatial dimensions while retaining meaningful spatial context.
- Fully Connected Layer:
  - Final classification layer replaced with a Linear layer of 2048 → 4 neurons.
  - Outputs four prediction logits corresponding to:
    - Splicing
    - Copy-Move
    - Inpainting
    - Face Manipulation

- Output Layer: Uses SoftMax activation (via CrossEntropyLoss) for multi-class classification.

B. Segmentation Architecture

For manipulation localization, we implemented the DeepLabV3+ architecture leveraging a ResNet-50 backbone pre-trained on ImageNet.

- Input Layer: Accepts RGB images resized to 256 × 256 × 3.
- Encoder Backbone (ResNet-50):

    - Acts as the feature extractor capturing multi-scale contextual patterns.
    - Utilizes residual learning to enable deeper representation and mitigate vanishing gradient issues.
- Atrous Spatial Pyramid Pooling (ASPP):
    - Extracts multi-scale semantic understanding using parallel atrous convolutions.
    - Enhances detection of irregular, fine-grained tampering boundaries.
- Decoder Head (DeepLabV3+):
    - Refines object boundary precision through upsampling and feature fusion.
- Output Layer:
    - A single-channel convolutional layer generates a pixel-wise probability map.
    - Output mask uses BCE With Logits during training, where 1 identifies manipulated pixels while 0 indicates authentic content.

This two-stage design allows the system to first detect manipulated images and then precisely locate the forged parts.

5.3 Hyperparameters Configuration

A set of optimized hyperparameters were defined for both the classification and segmentation models with considerations for GPU memory availability, dataset size, and convergence behavior.

A. Classification Model - ResNet-50 (Tampering Type Detection)

Hyperparameter	Value
Input Image Size	256 × 256 × 3
Batch Size	128 (split automatically across GPUs)
Learning Rate	1e-4 (later reduced for fine-tuning)
Optimizer	Adam
Loss Function	CrossEntropyLoss
Epochs	10
Weight Initialization	Pretrained on ImageNet
Data Loader Workers	16 workers
Seed	42
Device	CUDA (Multi-GPU Data Parallel)

Table 2: Hyperparameter description of the classification model

Rationale:
A moderate learning rate ensured stable fine-tuning over the pretrained ResNet-50 backbone. Batch size of 128 maximized GPU utilization while still maintaining memory efficiency. CrossEntropyLoss was chosen for multi-class prediction across four tampering manipulation categories.

B. **Segmentation Model — DeepLabV3+ (Pixel Mask Localization)**

Hyperparameter	Value
Input Image Size	256 × 256 × 3
Batch Size	64
Learning Rate	1e-4
Optimizer	Adam
Loss Function	BCEWithLogitsLoss + DiceLoss
Epochs	25
Weight Initialization	Pretrained DeepLabV3+ with ResNet-50 backbone
Data Loader Workers	16 workers
Device	CUDA (Multi-GPU Data Parallel)

Table 3: Hyperparameter description of segmentation model

Rationale:
Segmentation training required a lower batch size due to the computational cost of pixel-level prediction. BCEWithLogitsLoss was preferred to handle binary mask prediction efficiently. A longer training schedule (25 epochs) allowed gradual refinement of spatial boundaries.

5.4 Training Strategy

After preprocessing, the dataset will be divided into training (70%), validation (15%), and testing (15%) subsets.

- **Data Preprocessing:**
  All images are resized to a fixed dimension (256×256), normalized, and augmented using random flips, rotations, and brightness changes. For segmentation, corresponding masks are binarized and resized using nearest-neighbour interpolation.
- **Model Training:**
  Both models are trained using the Adam optimizer and Binary Cross-Entropy (BCE) loss function. For segmentation, an additional Dice Loss are combined with BCE to improve mask accuracy. Early stopping and learning rate scheduling are used to avoid overfitting.
- **Implementation Tools:**
  The models are trained using TensorFlow/Keras or PyTorch, and training is performed on GPU using Jupyter Notebook.

6.0 Results/Major Outcomes

This section presents the quantitative performance of the models based on standard evaluation metrics.

6.1 Quantitative Analysis

Classification Model: The classification model was evaluated using Accuracy, Precision, Recall, and F1-Score.

Metric	Score Achieved	Description
Accuracy	0.902	Overall correctness of predictions.
Precision	0.909	Proportion of predicted forgeries that were actually forged.
Recall	0.902	Proportion of actual forgeries correctly detected.
F1-Score	0.905	Harmonic mean of precision and recall.

Table 4: Performace Metrices of the Classifiction Model

Segmentation Model: The localization performance was measured using Intersection over Union (IoU) and Dice Coefficient.

Metric	Score Achieved	Description
IoU	0.6220	Overlap between predicted and actual forged regions.
Dice Coefficient	0.6689	Similarity between predicted and true masks.

Table 5: Performance Metrices of the Segmentation Model

6.2 Visual Outcomes

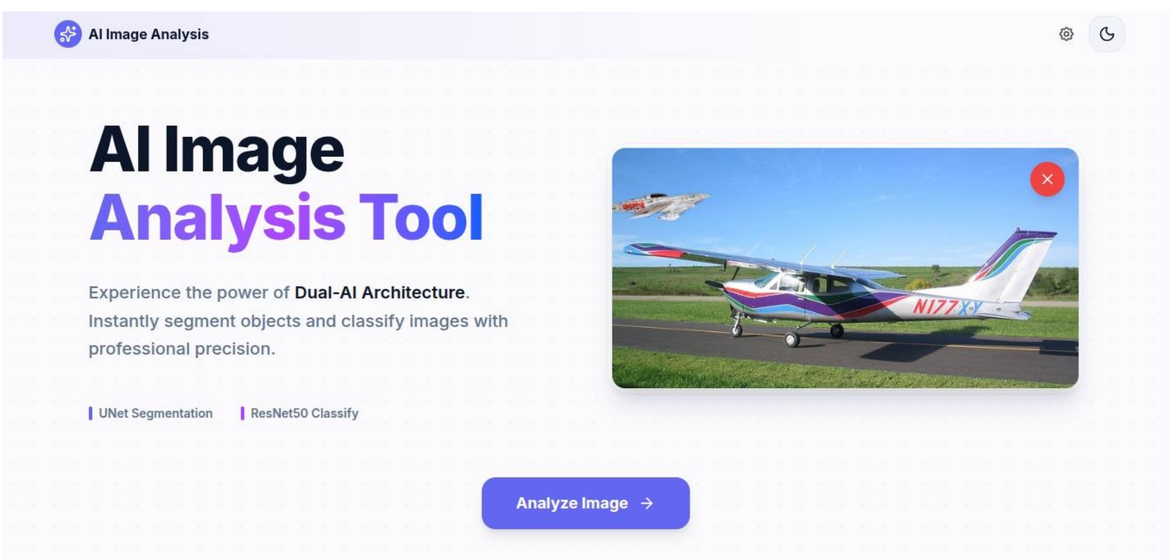

Figure 6: Homepage Interface

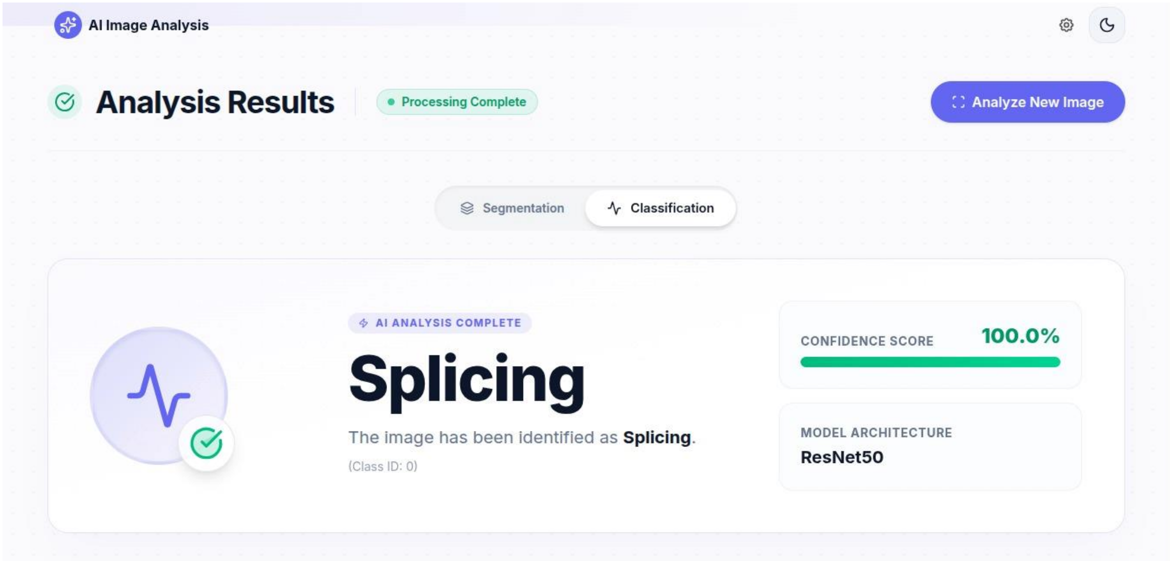

Figure 7: Classification Output Interface

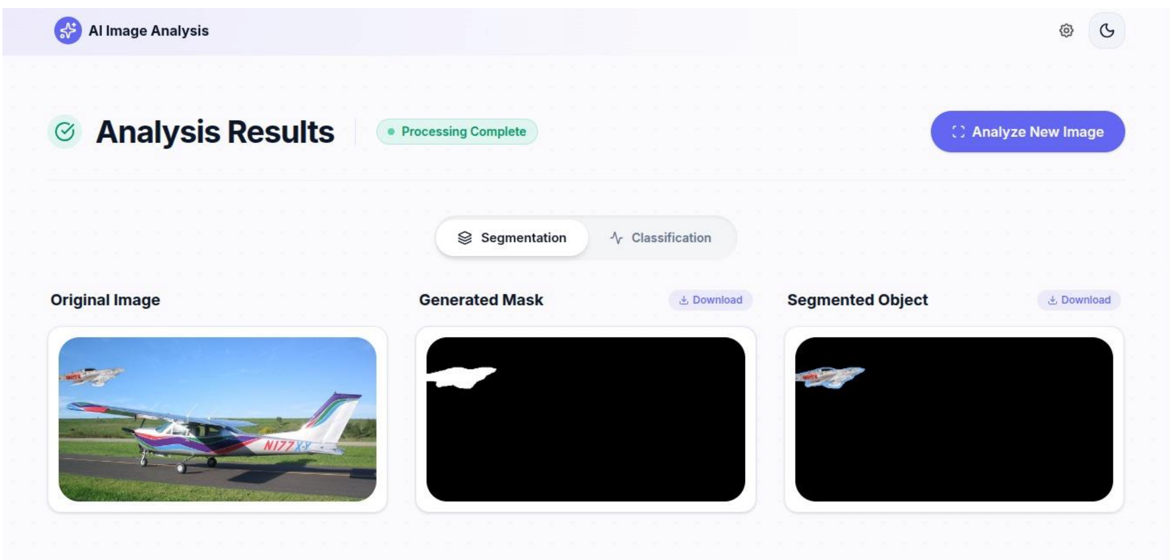

Figure 8: Segmentation Output Interface

7.0 Conclusion

This project successfully designed, implemented, and evaluated a comprehensive deep learning framework for Image Forgery Detection and Localization. By leveraging the large-scale DEFACTO dataset, the study addressed the complex challenge of distinguishing between authentic and manipulated content across multiple forgery types, including splicing, inpainting, and face manipulation.

The quantitative analysis highlights the robustness of the proposed two-stage architecture. The ResNet-50 classification model proved highly effective in identifying tampering types, achieving a commendable 90.2% accuracy. Furthermore, the DeepLabV3+ segmentation network succeeded in generating binary masks that localize manipulated pixels with an IoU of 0.622, providing essential spatial interpretability often missing in standard binary classifiers. While the segmentation results indicate the inherent difficulty of detecting fine-grained boundaries in high-quality forgeries, the system demonstrates strong potential for practical application.

Ultimately, this project bridges the gap between detection and explanation. By providing both a classification verdict and a visual mask of the tampered region, the developed system serves as a foundational tool for digital forensics experts, aiding in the fight against visual misinformation and restoring trust in digital media.

8.0 References

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Project Code Link:

https://github.com/UbedullahShaikh/image-manipulation-backend
https://github.com/UbedullahShaikh/image-manipulation-frontend